

The past, present and future of de novo protein design

<https://doi.org/10.1038/s41586-026-10328-7>

Received: 16 September 2024

Accepted: 20 February 2026

Published online: 29 April 2026

 Check for updates

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With deep-learning-powered advances in protein design methods, there is an ongoing paradigm shift in protein engineering from random selection to intentional computational design methods. Here we describe the current state of de novo protein design. While there is still room for improvement in success rates and activities, the long-standing challenges of designing new protein structures, assemblies and protein binders are close to being solved. The key current questions in these areas are not how to design, but what to design, and open-source design methodology such as RFdiffusion and ProteinMPNN together with protein structure prediction tools enable biochemists and molecular biologists to broadly explore possible applications. There has also been considerable progress in the de novo design of small-molecule target binders, enzymes and multistate protein systems. Current challenges for methods development include design of catalysts for reactions with high energy barriers and, more generally, design of switches and nanomachines that integrate binding, conformational change and catalysis. Over the next five to ten years, we anticipate the design of sophisticated protein nanomachines and materials with functionality ranging far beyond that generated during natural evolution for a wide range of applications in medicine, technology and sustainability.

The proteins in nature carry out a dazzling array of functions that underlie the remarkable capabilities of living things. For example, proteins mediate the template-directed synthesis of DNA and RNA, the conversion of solar energy into chemical-bond energy in photosynthesis, and the complex cell–cell interactions and signalling cascades in multicellular organisms. The sophisticated functionality of natural proteins evolved over hundreds of millions of years of natural selection to solve the problems that were critical during biological evolution. However, humans face new problems today: we live longer, so diseases such as cancer and Alzheimer's have become major challenges, and we are heating up and polluting the planet. The considerable functionality and versatility of the proteins in nature suggests that new proteins could provide solutions to many of these problems, and that such new proteins could arise over millions of additional years of evolution and natural selection. However, we do not have this time to wait, and there has therefore been considerable interest in generating proteins with new functions.

This Review describes the current state of de novo protein design: building new proteins with new functions from the ground up rather than by repurposing existing native proteins. De novo protein design is now moving so fast that researchers outside the field—and even some practitioners—do not have a good understanding of what problems are solved and what are not, and the problems in medicine and technology for which protein design is now, or will likely soon be, one of the best approaches. This Review is not intended to be comprehensive as space considerations do not allow for coverage of all of the varied work on de novo protein design over the past 20 years^{1–4}, nor the explosion of machine learning (ML) methods development in the past 5 years⁵. We also do not cover progress in protein engineering generally; for example, enzyme engineering by directed evolution⁶ and design of metal-directed assemblies using native protein building blocks⁷. Here we instead aim to provide the reader with a high level of understanding in the very rapidly increasing capabilities of de novo protein design, often using examples from our laboratory

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Review

for illustration. Our goals are threefold: first, for the general reader, to provide an overview of the progress in this area; second, for scientists outside of the field, to encourage the application of protein design methods to their research problems; and third, for methods developers, to encourage focus on unsolved rather than largely solved design problems.

Protein design challenges

The major classes of challenges that de novo protein design has sought to solve are outlined in Fig. 1a. In roughly the order that they were tackled, these are the design of monomeric protein folds and multiprotein assemblies (Fig. 1), proteins that bind to other proteins (Fig. 2), small-molecule binders, sensors and enzyme catalysts (Fig. 3), and multistate functional biomolecular systems (Fig. 4). In almost all cases, both the sequences and structures that we present are unrelated to those of natural proteins. We first provide an overview of protein design methodology, and then review the current progress in each of these areas.

Protein design methodology

From the beginnings of computational protein design in the 1990s until around 6 years ago, computations were carried out using physical chemistry-based models starting from Anfinsen's hypothesis that the folded state of a protein is at the global free-energy minimum for its sequence⁸. De novo protein design software such as Rosetta⁹ focused on solving three broad classes of problems for any specified design challenge: first, sampling new protein backbone structures that could address the challenge; second, assigning sequences to these backbones that direct folding to the desired structure; and third, ranking designs based on their computed energies. In energy-guided flexible backbone design trajectories, such as those used to generate the Top7 protein¹⁰ with a new fold and sequence, each step involved all three problem classes: backbone sampling, sequence design and energy calculation. To solve the energy calculation challenge, atomic force fields were developed that attempted to calculate the energy of a protein structure given the positions of all of its atoms: the van der Waals, hydrogen bonding, electrostatic, torsional and solvation energies were computed, weighted and summed. The force field used in Rosetta resembles the molecular mechanics force fields used for molecular dynamics simulations, but with an explicit representation of the geometric dependence of hydrogen bonding, which was found to be particularly important at protein interfaces. For the sequence design challenge, sampling methods were developed to generate low-energy sequences for a given structure, starting with the important work demonstrating that new sequences could be found for natural protein structures¹¹.

Starting in 2020, there has been a shift from such physics-based methods to ML-based approaches. These new methods have been enabled by the increasing accuracy of protein structure prediction achieved by deep-learning methods such as AlphaFold^{12,13} and RoseTTAFold^{14,15}, which provide a mapping from sequence space to structure space. In contrast to the physically based methods, which seek to generate very-low-energy sequence–structure pairs, the ML approaches seek to generate high-probability pairs. These are not synonymous, as energy calculations consider only a specific structure for a given sequence and are blind to the existence of other possibly lower-energy structures. Probability calculations, on the other hand, must implicitly consider all other possible structures, as a structure can only have high probability if other possible structures have low probability¹⁶. By training on the very large set of known protein structures, ML methods learn to accurately estimate these probability distributions. The advances in deep-learning methods for structural biology^{5,17} are due to not only the development of improved ML models and the availability of computing

resources to train them but also, importantly, the 60 years, billions of dollars of investment and dedicated work of tens of thousands of scientists to experimentally determine protein structures and collect them in the Protein Data Bank (PDB)¹⁸.

A first approach to using ML protein structure prediction methods for de novo protein design took advantage of prediction confidence measures generated by the networks along with the predicted structures^{19–21}. Sampling is carried out in sequence space and, at each step, the structure is predicted and sequence changes that increase prediction confidence are favoured. Many steps of such optimization in sequence space lead to sequences that are predicted with high confidence to have a particular structure. Experimental characterization of such sequences, generated first with an early structure prediction method called trRosetta²², and later with the more accurate AlphaFold2^{23,24}, demonstrated that sequences generated in this way did in some cases fold to the corresponding structures. By analogy to related approaches for image generation²⁵, this approach is referred to as hallucination or activation maximization—the latter because the solutions are found by directly optimizing over the inputs. However, this approach had two disadvantages: first, as at each step for evaluation a structure prediction calculation is necessary, it can be quite time consuming; and second, the sequences generated can be physically implausible. Hallucination approaches have been improved considerably by allowing mixtures of amino acids at each position during sampling²⁴.

Many of the most powerful and versatile approaches for protein design use generative artificial intelligence (AI) algorithms that were first developed for computer vision. RFDiffusion²⁶ and Chroma²⁷ were trained to generate protein backbones based on problem-specific input in a manner analogous to methods like DALL-E²⁸, which generates images based on a text prompt²⁵. In the latter case, images from the internet are noised to different extents, and a network is trained to remove the noise and regenerate the original image. New images can then be generated by starting from random noise, and repeatedly applying the network to progressively remove the noise; the resulting images look realistic, but are completely made up. In an analogous manner, RFDiffusion is trained to denoise partially noised versions of the structures in the PDB and, after training, new protein structures can be generated by progressively denoising starting random noise distributions (Fig. 1b). These approaches essentially model the underlying probability distribution of the training data, and generate new samples drawn from this distribution^{29–31}. During training, additional information can be provided with a subset of the noised samples; this enables the conditioning of the denoising diffusion process at inference to generate proteins with new functions. For example, RFDiffusion denoising trajectories in the presence of a small-molecule target generate new proteins predicted to bind to this target.

Many widely used ML design methods^{24,32–37} were developed in protein design rather than in ML research groups, probably because the overall structure of the problem, and the problems to be solved directly, carried over from the previous era of physically based models. The diffusion approaches carry out backbone sampling, ML sequence design methods^{38–40} such as ProteinMPNN generate sequences for fixed backbones and structure prediction replaced energy-based ranking approaches⁴¹. Once new sequences were generated, the same experimental evaluation approaches could be directly applied, involving the synthesis of genes encoding the designs, expression and purification of the designed proteins from *Escherichia coli*, and assays to assess whether the desired function had been obtained (Fig. 1a). Methods development was directly coupled to experimental testing. For example, it was found that, in hallucination approaches using AlphaFold2, the generated reasonable sequences did not actually fold to the intended structures, but using sequence design methods, such as ProteinMPNN, to generate sequences for the hallucinated backbones yielded correctly folding proteins^{23,42}. Models that simultaneously

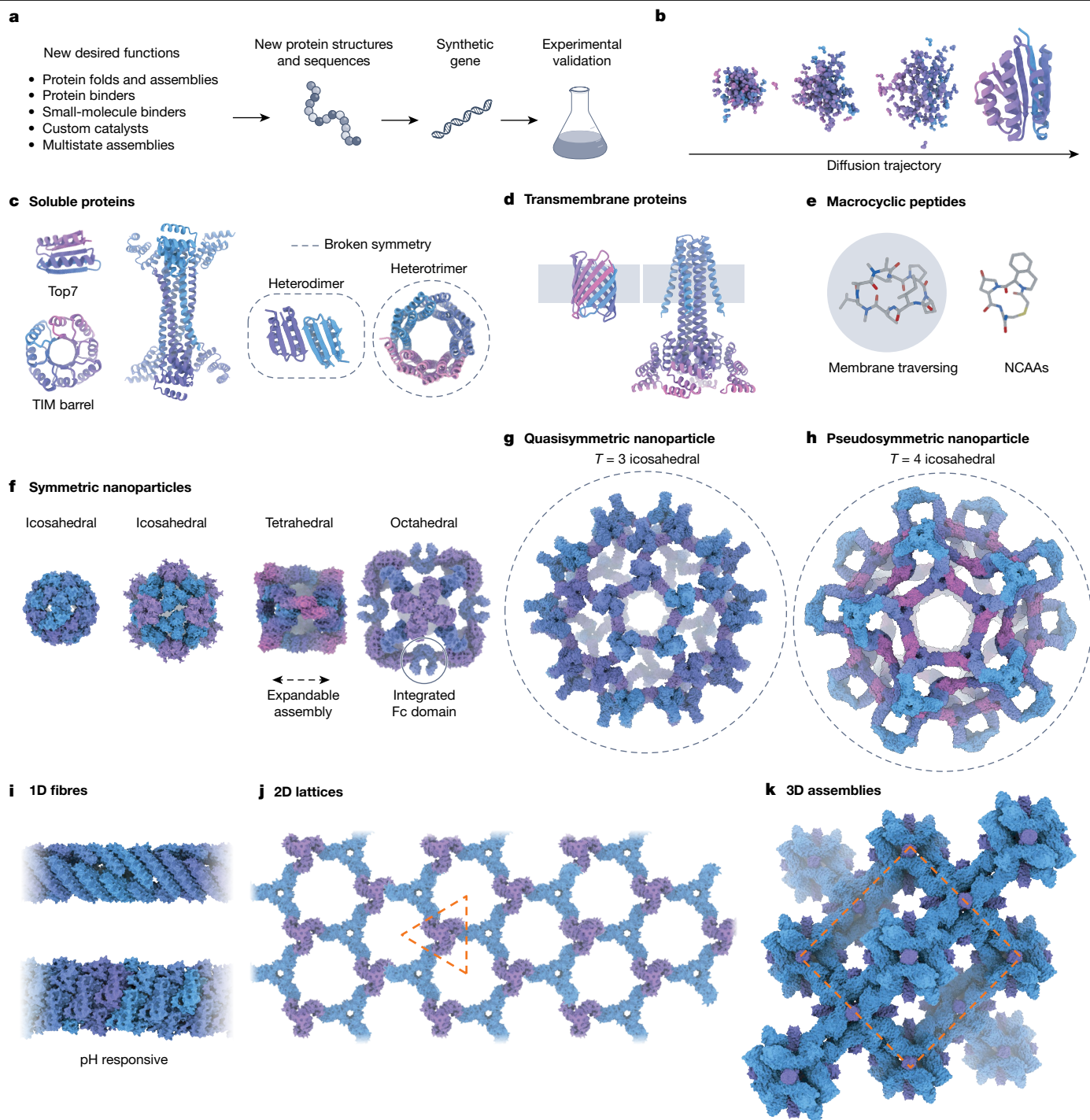


Fig. 1 | Overview of protein design challenges and design of protein folds and assemblies. **a**, Workflow for the computational design of proteins with new desired functions. **b**, An example RFdiffusion trajectory illustrating the generation of a protein monomer. **c**, De novo designed globular proteins with high stability. PDB IDs: 1QYS (Top7)¹⁰, 5BVL (TIM barrel)⁵⁹, 6WMK (heterodimer)¹⁰¹ and 8EOO (heterotrimer)¹⁰³. **d**, Designed transmembrane proteins. PDB IDs: 6X1K (left, β -barrel)⁷⁴ and 6B85 (right, transmembrane tetramer)⁷¹. **e**, Designed membrane-traversing macrocyclic peptides^{83,84} (left) and oligoamides with noncanonical amino acids (NCAs, right)^{83,84}. **f**, Designed nanoparticles with tetrahedral⁹², octahedral⁸⁹ and icosahedral symmetry^{90,91}.

g, Designed two-component nanoparticles with quasisymmetry. T , triangulation number. PDB ID: 9OP9. **h**, Designed four-component nanoparticles with pseudosymmetry^{106,107}. **i**, Designed one-dimensional fibres. PDB IDs: 8EOV (upper)¹⁰⁹ and SUBG (lower)¹¹⁰. **j**, A designed reconfigurable 2D lattice⁹³. **k**, Designed 3D crystalline assemblies¹¹². Designed monomers (**b–d**) are shown as a cartoon representation, coloured by a rainbow gradient from purple to blue according to the C α order. Designed assemblies (**f–k**) are shown as surface representations, with identical components coloured uniformly to highlight their symmetry and structural organization.

generate protein sequences and structures aim to capture the intrinsic coupling between folding and function^{43–46}.

Inspired by the remarkable success of natural language models such as BERT⁴⁷ and GPT⁴⁸, researchers have developed ML models capable of

understanding and generating protein sequences. Early efforts^{49,50} primarily focused on masked language models that were trained to predict the identities of randomly masked amino acids and thereby learn statistical and evolutionary patterns from large protein sequence databases.

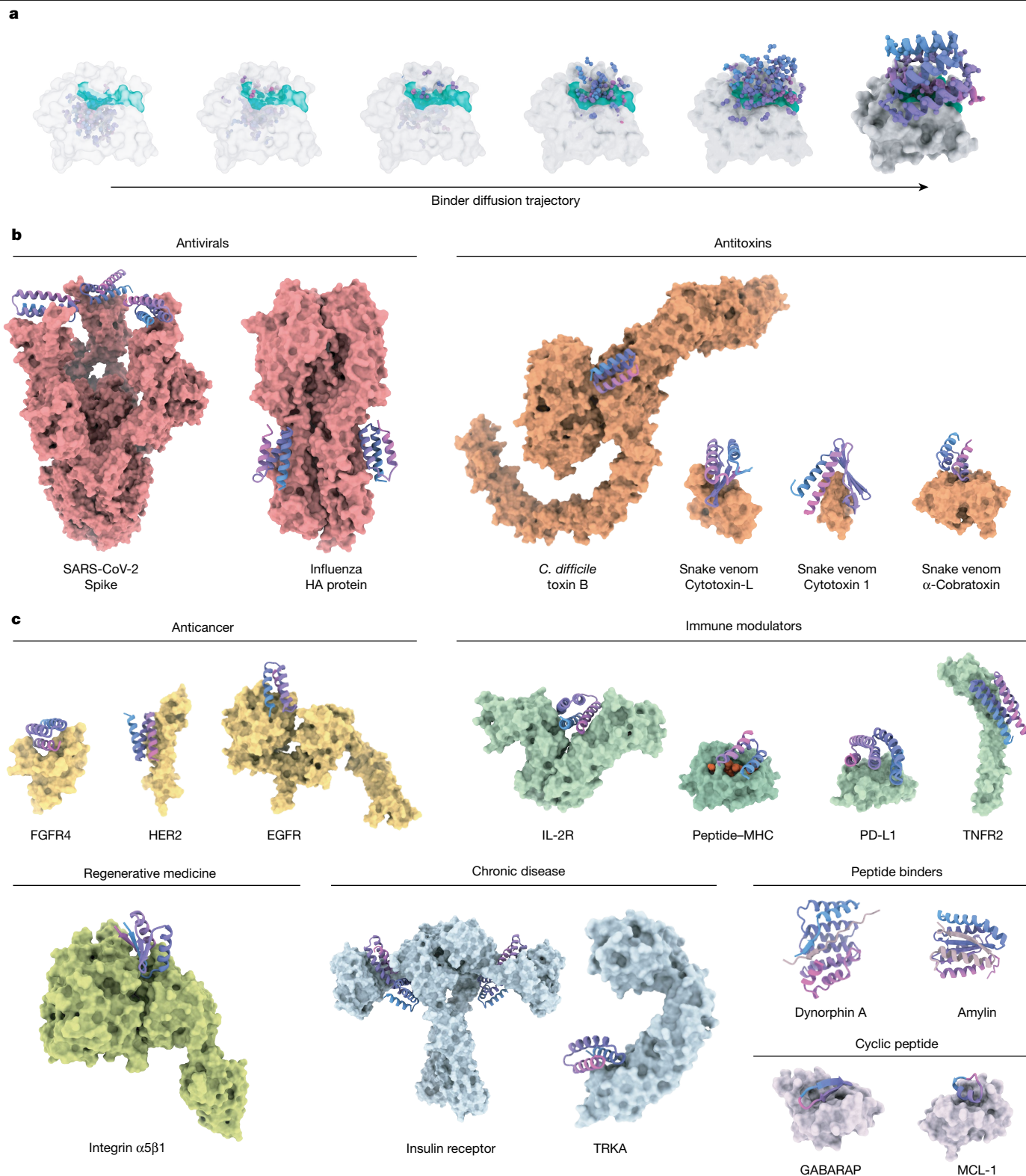


Fig. 2 | Design of binding to protein targets. a, An example RFdiffusion trajectory illustrating the generation of protein binders designed to target the specified binding site (green surface) on a receptor (grey surface). **b**, De novo designed protein binders for antiviral (left) and antitoxin (right) applications. PDB IDs: 7JZM (SARS-CoV-2 Spike)^{140,141}, 7RDH (influenza A virus haemagglutinin (HA))¹³⁹, 9BK5 (cytotoxin-L)¹¹⁹ and 9BK6 (cytotoxin 1)¹¹⁹. **c**, De novo designed protein binders for diverse therapeutic applications, with the receptor names

indicated below. PDB IDs: 7TYD (FGFR4), 6DG5 (IL-2R), 9UC8 (TNFR2), 7N3T (TRKA), 9CKV (integrin $\alpha 5 \beta 1$)²⁴³ and 9DN6 (insulin receptor)²⁴⁴. Designed protein binders (a–c) are shown as cartoon representations, coloured with a rainbow gradient from purple to blue according to the C α order. Native receptors are depicted as a surface representation and are coloured based on their application category.

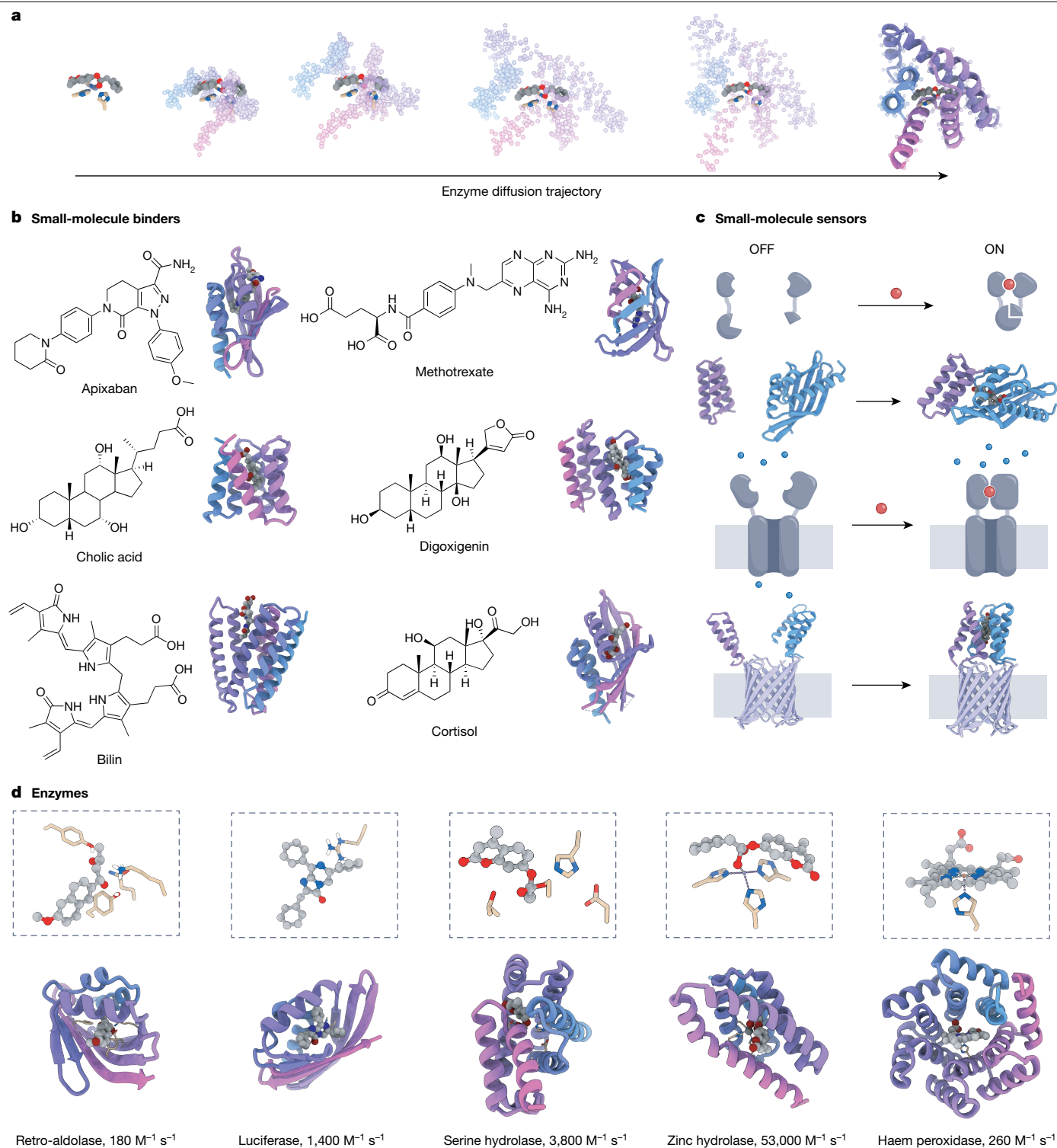


Fig. 3 | Design of small-molecule binders and enzymes. a, A representative example of an RFdiffusion2 enzyme design trajectory starting from a model of an ideal active site consisting of the reaction transition state surrounded by amino acid side chains mediating catalysis (left, in this case, a metallohydrolase with three histidines chelating a zinc ion). **b**, De novo designed binders targeting the small molecules apixaban¹⁷⁹, methotrexate¹⁸⁰, cholic acid¹⁸⁰, digoxigenin¹⁴, bilin¹⁴ and cortisol¹⁷⁹. PDB IDs: 8VFQ (apixaban binder)¹⁷⁹, 8VEI (cholic acid

binder)¹⁸⁰ and 8UQF (cortisol binder)¹⁷⁹. **c**, Schematic and design models of de novo designed small-molecule sensors using CID¹⁷⁹ and a designed nanopore¹⁸⁰. **d**, Enzyme active sites and de novo designed enzymes with corresponding $k_{\text{cat}}/K_{\text{M}}$ values: retro-aldolase²⁰⁵, luciferase²⁰⁴, serine hydrolase (PDB: 9MRB)²⁰⁶, zinc hydrolase (PDB: 9PYL)²¹⁴ and haem peroxidase (PDB: 8C3W)²⁰⁸.

To explicitly enable sequence generation, a second class of protein language models^{51–53} used an autoregressive formulation, predicting the next amino acid in a sequence and therefore supporting de novo sequence generation and family-conditioned design, particularly

after fine-tuning on specific protein families or functional motifs. Discrete diffusion-based approaches^{54–56} learn to unmask sequences or tokens from noise to achieve improved global coherence. Multi-modal protein language models³⁷ and sequence–structure codesign

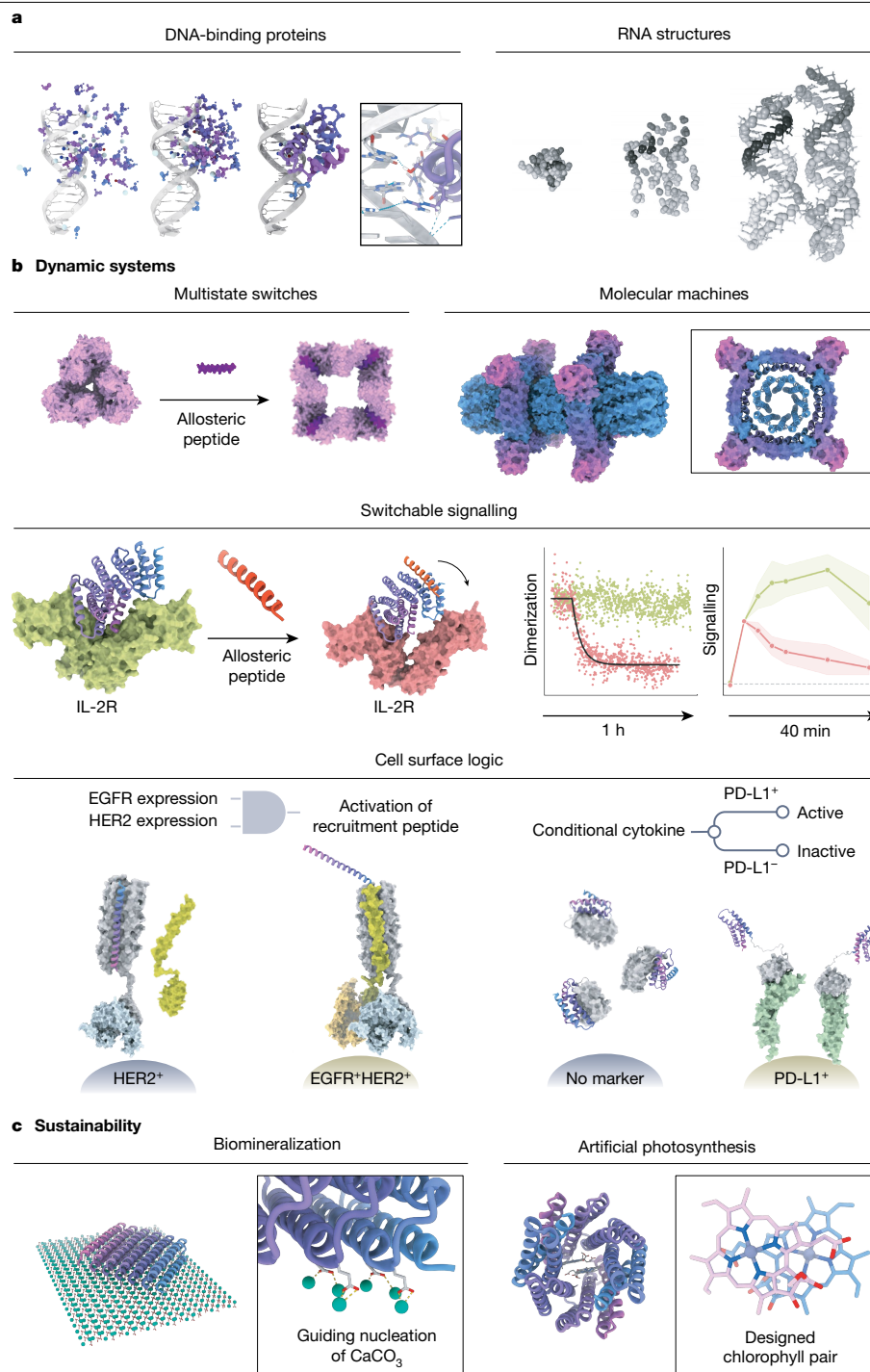


Fig. 4 | Design of proteins with new functions. **a**, Left, an example RFdiffusion trajectory illustrating the generation of proteins designed to bind DNA²¹⁹. Right, an example RFdiffusion-poly trajectory showing the generation of an RNA tertiary structure²²⁹. **b**, Top left, design of allosterically switchable cyclic assemblies²²⁹. Top right, design of mechanically coupled axle-rotor protein assemblies⁷⁹. Middle, design of facilitated dissociation to control cytokine signalling duration²³⁰. The kinetic traces on the right are with (orange trace) or without (green trace) allosteric peptide. Bottom left, design of AND logic through localizing the LOCKR cage (grey surface) and key (green surface) to the surface of the same cell (AND logic), which uncages a peptide (rainbow gradient from purple to blue, cartoon), recruiting effector proteins²²⁴. Bottom right, design of a conditionally active IL-21 mimic (rainbow gradient

from purple to blue, cartoon) that is uncaged by PD-L1 binding. The functional interfaces of the PD-L1 binder and the IL-21 mimic are docked together forming a caged state. Owing to the higher affinity of the PD-L1 interaction, engagement with PD-L1 disrupts this interface, leading to uncaging of the IL-21 mimic and rescue of its activity. **c**, Left, a designed protein template guides the heterogeneous nucleation of CaCO_3 ²³². Right, a designed homodimeric protein housing excitonically coupled chlorophyll special pairs²³⁶ (PDB: 7UNI). Designed monomers (**a–c**) are depicted as a cartoon representation, coloured by a rainbow gradient from purple to blue according to the $\text{C}\alpha$ order. Designed assemblies (**b** (top)) are shown as a surface representation, with identical components coloured uniformly to highlight their symmetry and structural organization.

methods⁴⁵ jointly model sequence, structure and function, enabling the simultaneous generation of protein sequences and structures. It remains to be seen how well such methods perform on current design challenges involving the accurate positioning of multiple sidechain atoms, for example, in an enzyme active site.

In the following sections, we describe the progress in the major protein design challenges. In each case, we provide an RFdiffusion trajectory to illustrate how generative AI methods are currently being used to solve the problem, a survey of the problems solved to date and an indication of current challenges in the area for the field.

Protein folds

De novo protein design first focused on the fundamental inverse folding problem: the design of amino acid sequences that fold to new monomeric protein structures. Methods were developed for the identification of novel sequences for existing folds, the diversification of existing folds and the creation of entirely non-native folds. These efforts have illuminated protein folding and design principles, and provide building blocks for generating proteins with diverse and sophisticated functionalities. It was shown in 1988 that new proteins could be designed by generating sequences encoding amphipathic α -helices⁵⁸; in 1997, it was shown that new protein sequences that adopt already existing folds can be identified by computational sidechain packing methods¹¹. Following these pioneering advances, it was shown that new globular protein structures could be designed using the Rosetta structure prediction methodology to sample new protein backbones and sequence design to identify low-energy sequences for these backbones¹⁰. The very close agreement between the crystal structure of the designed protein (Top7) and the computational design model showed that new proteins could be designed with atomic accuracy. Over the 20 years since Top7, a very wide variety of new structures have been designed and experimentally validated, some with folds resembling those in nature, such as TIM barrels⁵⁹, helical repeat proteins⁶⁰ and β -barrels⁶¹ (Fig. 1c) and many others with new folds^{62–67}. While new ML methods allow for greater control and novelty in designing structures and folds^{26,42,68–70}, this challenge is largely solved.

By incorporating hydrophobic surfaces driving membrane insertion, transmembrane proteins can now be de novo designed with high accuracy (Fig. 1d). Both α -helical^{71–73} and β -sheet^{74,75} transmembrane pores and channels^{76,77} have been designed that enable ions and other compounds to traverse the membrane, along with conformational switching zinc transporters⁷⁸. Single-molecule DNA sequencing has relied on naturally occurring β -barrel pores; de novo design now enables the generation of pores with custom sizes and linings⁷⁹, which could enable a wide range of new nanopore-based technologies, including protein sequencing and generalized sensing.

Computational design has been extended beyond the 20 natural amino acids to unnatural side chains and backbones⁸⁰. Owing to the challenges of using the natural protein translation machinery with many unnatural amino acids, there has been particular progress with short peptides and macrocycles that can be synthesized chemically. Rosetta design methods have been generalized beyond non-natural amino acids and backbones by extending force-field parameterizations and sampling methodologies to a much wider range of chemical structures. Seven-to-14-residue backbone cyclized peptides (Fig. 1e (left)) composed of L- and D-amino acids have been designed that adopt well defined structures in solution⁸¹, bind to protein targets⁸² and are membrane permeable and orally bioavailable⁸³. These methods have been further generalized to design minicycles with 4–5 residue oligoamide building blocks⁸⁴ (Fig. 1e (right)). All-atom modelling approaches^{85–87}, for example, RFdiffusion-all atom, have been extended to cyclic peptide systems into which both D- and L-amino acids can potentially be incorporated. Thus, approaches originally developed for protein design can be applied to the design of small drug-like molecules.

Box 1

History of designed protein assemblies

The de novo design of symmetric protein assemblies started from human expert designed helical bundles in the late 1980s²⁴⁵ and progressed to diverse self-assembling nanomaterials in the 2010s, including one-dimensional protein filaments, two-dimensional arrays and three-dimensional nano cages and crystals. Before 2020, three main approaches were used for generating protein assemblies: (1) generation of backbone arrangements using parametric equations, initially developed for helical bundles with cyclic symmetries such as coiled coils^{246,247}; (2) rigid fusion of cyclic protein oligomers by aligning their internal symmetry axes with those of a desired architecture^{248–250}; (3) symmetric docking of cyclic oligomers as sequence-independent rigid bodies, followed by combinatorial sequence optimization at the newly formed interface to stabilize assembly^{251–253}. Sequences at the newly generated protein–protein interfaces or fusion junctions were designed using methods such as Rosetta to ensure specificity and stability. Since 2020, the development of deep-learning approaches, such as hallucination^{19,23}, reinforcement learning⁹¹ and RFdiffusion²⁶, has transformed the symmetric backbone generation process, enabling more extensive sampling of backbone conformations in the context of the desired symmetry. The improved backbone generation methods, coupled with fast and accurate fixed backbone sequence design (for example, ProteinMPNN⁴¹) and structure prediction methods (AlphaFold¹² and RosettaFold¹⁵) for evaluating whether the designed sequences assemble to the target structure, now enable the design of protein assemblies with complex folds and functions.

Protein assemblies

Incorporation of interfaces driving self-assembly has enabled design of a wide range of new protein assemblies⁸⁸ and nanomaterials with applications in vaccine development, signalling⁸⁹ and drug delivery. Cyclic oligomers and polyhedral nanocages with predefined sizes and porosities have been designed with a variety of methods, including Rosetta⁹⁰, symmetric hallucination²³, RFdiffusion²⁶ and reinforcement learning⁹¹. Customizable nanomaterials have been designed that use idealized building blocks that can be modularly expanded⁹² or combined⁹³ like Lego pieces, enabling straightforward blueprinting of a wide variety of architectures and symmetries (Fig. 1f, Box 1 and Supplementary Fig. 1).

Designed icosahedral nanoparticles have led to the first de novo designed, clinically approved medicine: the SKYCovione COVID-19 vaccine, which was recently authorized for human use^{90,94}. This approach is now being applied broadly to develop effective vaccines against diverse viruses⁹⁵, including a potential universal flu vaccine⁹⁶, which can be encoded as mRNA for efficient delivery⁹⁷. Incorporating additional functional motifs into protein particles opens a wealth of possibilities, including antibody cage-based platforms⁸⁹ that trigger specific therapeutic signalling, pH-responsive particle dissociation for delivery⁹⁸ and cryo-electron microscopy scaffolds that advance structural biology⁹⁹.

By breaking perfect point-group symmetry and allowing subunits to adopt distinct local environments, larger and more complex assemblies have become accessible (Fig. 1g,h). At the building-block level, the design of heterodimers and heterotrimers^{100–103} provides a starting point for constructing modular, multicomponent assemblies that have been used for protein logic and circuit design^{104,105}. Building on de novo heterotrimeric building blocks¹⁰³, pseudosymmetric

cages^{106,107} have been generated, greatly expanding both the maximum size (up to 100 nm in diameter using four components; Fig. 1h) and the functional diversity of possible protein materials. Through the design of local geometric frustration to induce global symmetry breaking, two-component quasi-symmetric particles have been designed, with potential applications as biological probes and delivery vehicles.

Unbounded one-, two- and three-dimensional designed nanomaterials have considerable potential for nanoscale patterning in both nanotechnology and synthetic biology. A wide variety of protein fibres has been designed¹⁰⁸, including extendable¹⁰⁹, pH-dependent¹¹⁰ and allosterically modulated¹¹¹ systems (Fig. 1i). Biologically active, two-component, two-dimensional arrays¹¹² induce cell surface protein clustering and polarization¹¹³. A reconfigurable 2D array (Fig. 1j) designed with RFdiffusion rapidly converts into cages when complementary binding partners are present⁹³. Three-dimensional protein crystals can now be robustly designed using a hierarchical design strategy¹¹⁴ following earlier successes for specific coiled-coil cases¹¹⁵. Designing crystalline protein frameworks that self-assemble under physiological conditions holds promise for a wide range of applications at the interface of biology and materials science (Fig. 1k).

The potential of computational protein design to create customized nanomaterials is rapidly catching up to the abilities of DNA nanotechnology. As these designed proteins can be genetically encoded and expressed in a range of living systems, they hold great promise for applications ranging from mRNA directed vaccines to structural, signalling and control units within cells.

Design of binders to protein targets

Binders to protein targets have considerable potential as therapeutics for blocking receptor–ligand and other interactions with potential advantages over antibodies in modularity, stability and size. Designed binders can also function as affinity reagents and diagnostics for purification and sensitive detection of protein targets. Generative AI methods¹¹⁶ such as RFdiffusion²⁶ and gradient-based hallucination approaches such as Bindcraft have been used to robustly generate binders against many protein targets¹¹⁷; other methods include Chroma²⁷, ProGen⁵² and PXDesign¹¹⁸. To illustrate how these generative AI methods for protein design work, a representative RFdiffusion binder design trajectory is shown in Fig. 2a: a random cloud of amino acid residues is placed around a user-specified site on the structure of a target of interest and, over the diffusion trajectory, these residues organize into a protein structure with high complementarity to the target. Complementarity and binding affinity can be refined by partial diffusion, incorporating a small amount of noise on the successful binder and then denoising; using this approach, designs with picomolar affinity have been obtained to targets as varied as snake venom toxins¹¹⁹ (Fig. 2b) and the TNF receptor¹²⁰. These computational methods can be further conditioned on the antibody fold to generate nanobody (VHH) and scFv binders against targets of interest^{121–125}. For RFdiffusion, the conformations of the CDR loops and the overall docks are freely sampled by the diffusion process. Directing RFdiffusion to generate designs that form extended β -sheets with target proteins (a strategy first performed using Rosetta¹²⁶) has considerably improved success rates against challenging polar targets¹²⁷. Computational design of macrocyclic peptides with cell-penetration abilities through computational programs such as HELM-GPT¹²⁸ and RFpeptides¹²⁹ in the presence of protein targets can now generate high-affinity drug-like binders (Fig. 2c).

Physics-based methods such as Rosetta and ML-based methods such as PepPrCLIP¹³⁰ and RFdiffusion can now target intrinsically disordered regions of proteins (IDRs)^{131–133} and peptides that have long been considered undruggable. The underlying design concept is induced fit—in contrast to folded targets, IDRs and peptides can be induced to adopt conformations that are particularly conducive to binding (for example,

with hydrophobic residues on one interaction surface). For example, in RFdiffusion trajectories targeting IDRs, the sequence of the target protein is kept fixed but the structure allowed to freely diffuse together with the generated binder. Designed binders to the oncogenes β -catenin and SS18–SSX1 can act as affinity handles for target degradation¹³⁰. Designed binders to segments of β -amyloid and tau—proteins that drive neurodegeneration-associated amyloids—potently suppress amyloid formation and have potential therapeutic use¹³⁴. Designed high-affinity binders to bioactive peptides and to fragments of natural proteins¹³⁵ are showing promise for detection and proteomics.

Together, the combination of Rosetta, RFdiffusion and ProteinMPNN have led to the design of experimentally validated binders to over 200 protein targets (Supplementary Table 1). Many more have been generated using the many other computational methods such as the hallucination-based BindCraft¹¹⁷. Although there is still room for improvement in the affinity and success rates, particularly for highly polar targets¹³⁶, the design of protein binders is approaching being a solved problem. Given the sustained success in designing high-affinity binders, the main question in the protein binder design area is no longer how to design binders, but what to design binders to and for what application. In the following sections, we review several major application areas.

Combating viruses

Small stable proteins have been designed to bind to and neutralize influenza^{137–139}, SARS-CoV-2^{130,140,141}, MERS-CoV¹⁴², the henipaviruses Nipah and Hendra, and respiratory syncytial virus either through fully de novo design^{137,139,141–143} or by incorporating the host receptor interface into a de novo scaffold^{140,141}. The focus has been on combating respiratory viruses owing to their pandemic potential, public health significance and susceptibility to neutralization through intranasal administration. In all of the above cases, the designed binders block viral infection with a potency comparable to or greater than antibodies, particularly after oligomerization to match the oligomerization state of the target viral glycoproteins^{142,144,145}, which also makes them more resistant to escape mutations^{142,144,145}. For influenza, SARS-CoV-2 and MERS-CoV, intranasal administration of designs provides potent prophylactic protection and/or therapeutic efficacy in animal models^{142,146,147}. The high thermostability of designed proteins obviates the need for a cold chain, and it could be possible to formulate a nasal spray incorporating designed antagonists to all major respiratory viruses that could be readily administered at home during the period of seasonal respiratory virus infection, or during pandemic outbreaks. In this area, the main challenge is not design methodology or even early-stage characterization of efficacy but, instead, the later-stage problem of how to advance this new modality into the clinic.

Neutralizing toxins

Small stable proteins have been designed to neutralize the botulinum neurotoxin B¹³⁷, *Clostridioides difficile* toxin B^{148,149}, the *Paenicitostidium sordellii* lethal toxin TcsL¹⁵⁰ and the major components of snake venom¹⁵¹. In vivo studies have demonstrated that the snake venom and toxic shock designs provide potent protection against lethal doses of toxin^{150,151}. The high stability and long shelf life of designed proteins could provide considerable advantages in cases like snake bites, which are prevalent in the developing world where cold chains may be less well established and current therapies are inadequate.

Receptor antagonists and agonists

Small stable proteins have been designed against a wide array of receptor tyrosine kinase extracellular domains (Fig. 2c and Supplementary Table 1). For receptor kinases, these designs function as potent antagonists as monomers, and can be converted into agonists by incorporation into homo- and hetero-oligomers that bring together multiple receptor subunits such that their intracellular kinase domains can

transphosphorylate. For GPCRs that are activated through intricate intramolecular rearrangements, the inactive or active conformation of the GPCR is provided as the target structure for protein design to create antagonists or agonists, respectively¹⁵². For cancer immunotherapy, binders have been designed against peptide-MHCs¹⁵³, which can be cancer neo-epitopes¹⁵⁴, cytokine receptor subunits, immune checkpoint proteins such as PD-1, PD-L1 and CTLA-4^{155,156} that potentially modulate immune cell signalling, and targets upregulated in cancer such as HER2. De novo designed analogues of IL-2¹⁵⁷ and IL-21¹⁵⁸ have shown extremely potent anti-tumour activity in animal models, and IL-2 analogues have advanced into human clinical trials. For regenerative medicine, binders have been generated against growth factor subunits and synthetic FGF agonists have been found to promote vascular development with more selective outcomes than native FGF¹⁵⁹. For autoimmune disease, high-affinity antagonists of major inflammatory cytokine receptors such as TNFR¹²⁰ and IL-23 have been designed; in the latter case, the high stability of designed proteins enabled oral delivery and amelioration of disease in an animal inflammatory gut disease (colitis) model¹⁶⁰. New agonists have been generated by linking together binders to receptor subunits that are not known to pair in nature, generating a wide variety of new signalling outcomes in immune cells and differentiating stem cells by bringing together the intracellular tyrosine kinase domains, which then cross-phosphorylate^{161–163}. The primary challenge here is not how to design binders to individual receptor subunits, but how to choose which pairs (or triplets) of receptors to bring together using this approach. De novo designed binders have been integrated into receptors^{164,165} such that target binding triggers downstream signalling, with potential use in adoptive cell therapies such as chimeric antigen receptor (CAR) T cells.

Trafficking and degradation

Designed proteins that bind to receptors that are endocytosed or transcytosed provide access to new therapeutic modalities for intracellular delivery and protein degradation. Designs against transferrin receptor^{126,166} and other receptors at the blood–brain barrier (such as sortilin¹⁶⁷) cross ex vivo blood–brain barrier models, and fusions could enable transport of binders to therapeutic targets in the brain. Several design strategies have been developed for driving rapid endocytosis and lysosomal targeting through receptors with different tissue distributions such as IGF2R, sortilin and ASGPR; these lysosomal trafficking stimulated designs are called EndoTags. Fusion of EndoTags to binders to disease-causing membrane and extracellular proteins leads to rapid clearance of the targets and improves efficacy of antagonistic antibodies in vivo¹⁶⁷. Methods such as PepPrCLIP¹³⁰ have successfully designed binders to various oncogenic proteins and attached E3 ubiquitin ligases to these binders for target protein degradation.

Small-molecule binders and sensors

De novo protein design can target small-molecule drugs, probes and metabolites, with applications ranging from in vivo reporters to diagnostics. Small-molecule binding proteins have been designed using physically based^{61,69,168–178} and deep-learning^{14,179–182} methods (Fig. 3b) for scaffold generation and binding sequence design¹⁸². The development of robust de novo protein scaffolds has enabled the design of binders for small molecules with diverse chemical properties^{173–175,178,183}. Generative AI models such as RFdiffusion have been extended to enable direct generation of protein structures around target small molecules^{184,185}. A representative RFdiffusion2 trajectory for enzyme design is shown in Fig. 3a, a similar approach is used to generate small-molecule binders.

Binders to therapeutics such as the heart disease drug digoxigenin and methotrexate have been designed and have potential as therapeutic sponges and sensors for monitoring drug levels (Fig. 3b). Designs against drugs of abuse such as fentanyl have been converted into

plant-based sensors by coupling binding to stabilization¹⁷⁰. Designs have been converted into chemically induced dimerization (CID) sensors by designing a second protein that interacts with the small-molecule–protein complex^{179,181} (Fig. 3c). Design of shape-complementary pseudocyclic scaffolds has enabled binding to polar and flexible compounds that have been incorporated into gated designed nanopores^{75,180} (Fig. 3b,c). For orthogonal control of cell signalling^{186–188}, CID systems^{172,181,189–192} have been designed that are triggered by clinically approved antiviral drugs or designed cyclic peptides.

Enzymes

The design of high-activity enzyme catalysts has been a long-standing goal of de novo protein design, both to test our fundamental understanding of catalysis and to generate new catalysts for reactions of current pressing concern such as plastic degradation. Early efforts started from a description of an ideal active site composed of a reaction transition state surrounded by key catalytic residues and sought to find geometric matches for this site in proteins of known structure. Catalysts were generated¹⁹³ for model chemical reactions such as retro-aldol¹⁹⁴ and Kemp elimination^{195,196} using this approach, but the activities were low and considerable directed evolution was required to approach the much higher activity levels of typical native enzymes^{197–200}. Structural characterization suggested that a shortcoming of these designs was the inability to precisely place protein functional groups around the transition states. This arises from both the complexity of native protein structures, which complicates the prediction of the effects of the sequence changes used to incorporate the active site, and the more general lack of atomic-level control over functional group placement. More robust scaffolds have improved success rates. Designed helical bundles with di-iron active sites have been specialized for different catalytic activities^{201–203}. Using a robust set of ML-generated scaffolds enabled the de novo design of high-activity luciferases²⁰⁴ (Fig. 3d).

Generative AI methods such as RFdiffusion2 now enable the direct construction of custom scaffolds to house any desired catalytic site, which has led to designed enzymes for a variety of chemical reactions^{14,26,193} (an example of a diffusion trajectory building up a scaffold around an enzyme active site is shown in Fig. 3a). Selection for preorganization of the active-site residues in catalytically competent conformations at each step in a reaction mechanism using an active-site ensemble-generating deep-learning method called PLACER²⁰⁵ considerably increased the design success rates for serine hydrolases²⁰⁶ using the classic catalytic triad plus oxyanion hole catalytic motif. The RFdiffusion buildup approach enables exploration of active site composition and geometries as the starting active-site description can be readily varied; in the serine hydrolase case, it was found that the inclusion of the full Ser-His-Glu triad and two oxyanion hole contacts led to higher activity, with catalytic efficiencies ($k_{\text{cat}}/K_{\text{M}}$) up to $2.2 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$. RFdiffusion has also been used to design proteins catalysing a retroaldol reaction²⁰⁷. Cofactors such as haem can be incorporated into de novo designs by constructing appropriate binding pockets^{14,208,209}; extension to unnatural cofactors should considerably expand the range of chemistries accessible to catalysis^{178,210–213} (Fig. 3d).

To enable the generation of protein structures around active sites specified at the level of catalytic side-chain functional groups (called atomic motifs) that directly interact with the transition state, and to avoid the need to specify where in the sequence these catalytic residues are placed, RFdiffusion2 enables direct structure generation around a set of atomic motif guideposts that specify the active-site geometry in a sequence agnostic manner²¹⁴ (see the diffusion trajectory in Fig. 3a). This, along with filtering for active site preorganization with PLACER, enabled the design of metallohydrolases with much higher activities ($k_{\text{cat}}/K_{\text{M}} > 10^4 \text{ M}^{-1} \text{ s}^{-1}$) compared with most previous computationally designed enzymes; the most active design had the greatest predicted preorganization²¹⁴.

With these advances, we anticipate that computational enzyme design will move beyond model chemical reactions to generate catalysts to solve outstanding current challenges in sustainability and biomanufacturing.

Outlook

De novo protein design is rapidly emerging as the state-of-the-art approach in many molecular engineering applications. For affinity reagent and therapeutic antibody generation, classic techniques such as animal immunization and random library selection will likely be supplanted by computational design, which is faster, cheaper and more precise. For multispecific and conditionally active therapeutics that act only at targeted locations in the body, de novo protein design enables more precise coupling between different interacting domains and could therefore provide more precise control than bispecific antibody constructs. In enzyme engineering, in which the current strategies rely on bioprospecting or directed evolution, de novo design enables the construction of custom active sites for new reactions and seamless incorporation of both natural and non-natural cofactors, but methodological improvements are still required to achieve high activities on less activated substrates.

With the growing power of computational design, many new challenges become accessible. In genome regulation, new methods for designing protein–nucleic acid interactions are opening avenues for next-generation gene therapy and control. Current technologies rely primarily on engineering natural systems, such as zinc fingers²¹⁵, transcription activator-like effectors²¹⁶ and CRISPR–Cas²¹⁷, which are constrained by the fixed geometry of their native scaffolds²¹⁸. More compact, customizable DNA-binding proteins with enhanced modularity and deliverability can now be designed using Rosetta and RFdiffusion approaches²¹⁹ (Fig. 4a (left)). Beyond DNA, RFdiffusion has been generalized to RNA and protein–RNA 3D structure design (Fig. 4a (right)); progress in this area could lead to new protein–RNA functional assemblies, expanding the frontier of synthetic biology.

Nature's molecular machines harness chemical fuel to power motion, synthesize molecules and maintain cellular quality control. Recent advances in binding and catalytic protein design now bring dynamic, multistate systems within reach, enabling transistor-like logic operations and providing building blocks for increasingly sophisticated molecular machines. For example, the LOCKR system integrates binding inputs into outputs such as bioluminescence^{220,221}, fluorescence^{221,222}, degradation²²³ or relocalization²²³, and can perform Boolean logic to direct CAR T cells to target specific surface marker combinations²²⁴ (Fig. 4b (bottom left)). Beyond LOCKR and small-molecule-modulated fusion proteins^{225,226}, hinge-like protein switches have been designed with two well-defined states toggled by an effector such as a peptide²²⁷ or calcium ion²²⁸. By coupling these hinges with reversible interaction modules, allosteric assemblies have been designed in which effector binding at a distant allosteric site changes the oligomeric state, leading to effector binding cooperativity²²⁹ (Fig. 4b (top left)). Allosterically controllable cytokines have been designed to dissociate rapidly from their receptor after addition of an effector, enabling control of the duration of signalling²³⁰ (Fig. 4b (middle)). Mechanically constrained nanoscale architectures with internal degrees of freedom⁷⁹, such as microscopic axles and rotors, have been designed; rather than locking into a single orientation, these self-assemble into rotary devices (Fig. 4b (top right)). Such mechanical coupling is a hallmark of machines and paves the way for nanoscale devices powered by solar energy or chemical fuels.

On the sustainability front, biomineralization in nature exemplified by bone, tooth and seashell formation offers inspiration for protein design approaches to control mineralization (Fig. 4c (left)). By creating de novo protein scaffolds that precisely interact with inorganic surfaces or lattice structures²³¹, it is possible to guide nucleation and growth; for example, programming calcite formation from calcium

carbonate for CO₂ fixation²³², and templating zinc oxide nucleation for semiconductor biosynthesis²³³. While natural mineralization primarily uses earth-abundant compounds like calcium carbonate and calcium phosphate, protein design has no such limitation and could enable bottom-up generation of new semiconductors and materials, complementing current top-down manufacturing methods. In renewable energy, protein design could accelerate breakthroughs in high-efficiency photosynthesis for faster-growing food crops, renewable solar fuel nitrogen fixation (Fig. 4c (right)). By designing simpler yet robust analogues of complex photosynthetic systems, it should become possible to incorporate near-infrared light into oxygenic photosynthesis^{234,235}, recreate the chlorophyll special pair at the reaction centre²³⁶ and self-assemble multinuclear metal clusters to facilitate multielectron reactions like water oxidation. Orchestrating these energy- and electron-transfer events could lead to new efficient, light-driven chemical processes, potentially surpassing natural photosynthesis in productivity^{234,235,237–239}.

Despite these advances and the considerable potential, several challenges must be addressed for de novo protein design to be fully transformative. Immunogenicity remains a key concern in therapeutic applications: while initial data indicate that many de novo proteins elicit relatively low immune responses, perhaps due to their high stability and solubility, further work is needed to minimize T cell epitopes and rigorously evaluate immunogenicity²⁴⁰. Large-scale protein manufacturing can also be a challenge, particularly for non-medical applications requiring bulk protein production; fortunately, de novo designed proteins are generally highly expressed inside cells and soluble, with high purification yields and high thermostability²⁴¹. As design efforts move beyond binding and enzymatic functions, new experimental assays and high-throughput screening will be critical for evaluating newly emerging activities. As with any new technology, there is the potential for misuse: the same computational tools capable of creating therapeutic or environmentally beneficial proteins could, in principle, be misused to design toxins²⁴² (nature has, however, already generated numerous lethal viruses and toxins, and the potential for protein design to protect humans from these probably far outweighs hypothetical dangers from new ones). On the legal side, the ease of sequence redesign raises questions regarding intellectual property within an evolving patent landscape.

In summary, de novo protein design is driving a paradigm shift in molecular engineering. While there is still room for improvement in success rates and activities, the fundamental challenges of designing new structures, assemblies and binding proteins are largely solved, and considerable progress is being made on catalysis, albeit still primarily for lower-energy-barrier reactions. Now that folding, assembly, binding and catalytic modalities are accessible to de novo protein design, the next frontiers involve more sophisticated functionality that combines two or more of these modalities. For example, the design of nanomachines that cycle through different conformational states driven by a chemical fuel, and therapeutics that go beyond antagonistic antibodies by not only binding to their target but also modifying or degrading it. The coming decade promises to be an exciting era of innovation as de novo protein design tackles major outstanding problems facing humanity and matures into a foundational tool across medicine, sustainability and biotechnology.

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Acknowledgements We thank our colleagues in the Baker Lab, the Institute for Protein Design and the Rosetta community for their contributions to much of the work described in this Review.

Author contributions The different sections were contributed by different author groups. D.B., D.J., G.B. and W.Y. contributed to the section on protein design methodology. D.B., A.C., S.W. and W.Y. contributed to the section on protein folds and assembly. D.B., J.Z.Z., X.W., T.S., M.A., R.R. and W.Y. contributed to the section on the design of binding to protein targets. D.B., G.R.L., L.A., I.K., S.P. and L.M. contributed to the section on small-molecule binders, sensors and enzymes. D.B., L.S., C.G., A.P., A.B., N.E. and W.Y. contributed to the Outlook section. E.H. and N.M.-B. contributed to Supplementary Table 1. S.W. contributed to Box 1. W.Y., G.R.L., J.Z.Z., S.W. and I.H. contributed to the illustration design and rendering. All of the authors reviewed the manuscript and provided feedback and revisions to their respective sections.

Competing interests The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10328-7>.

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Peer review information *Nature* thanks Martin Pacesa and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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